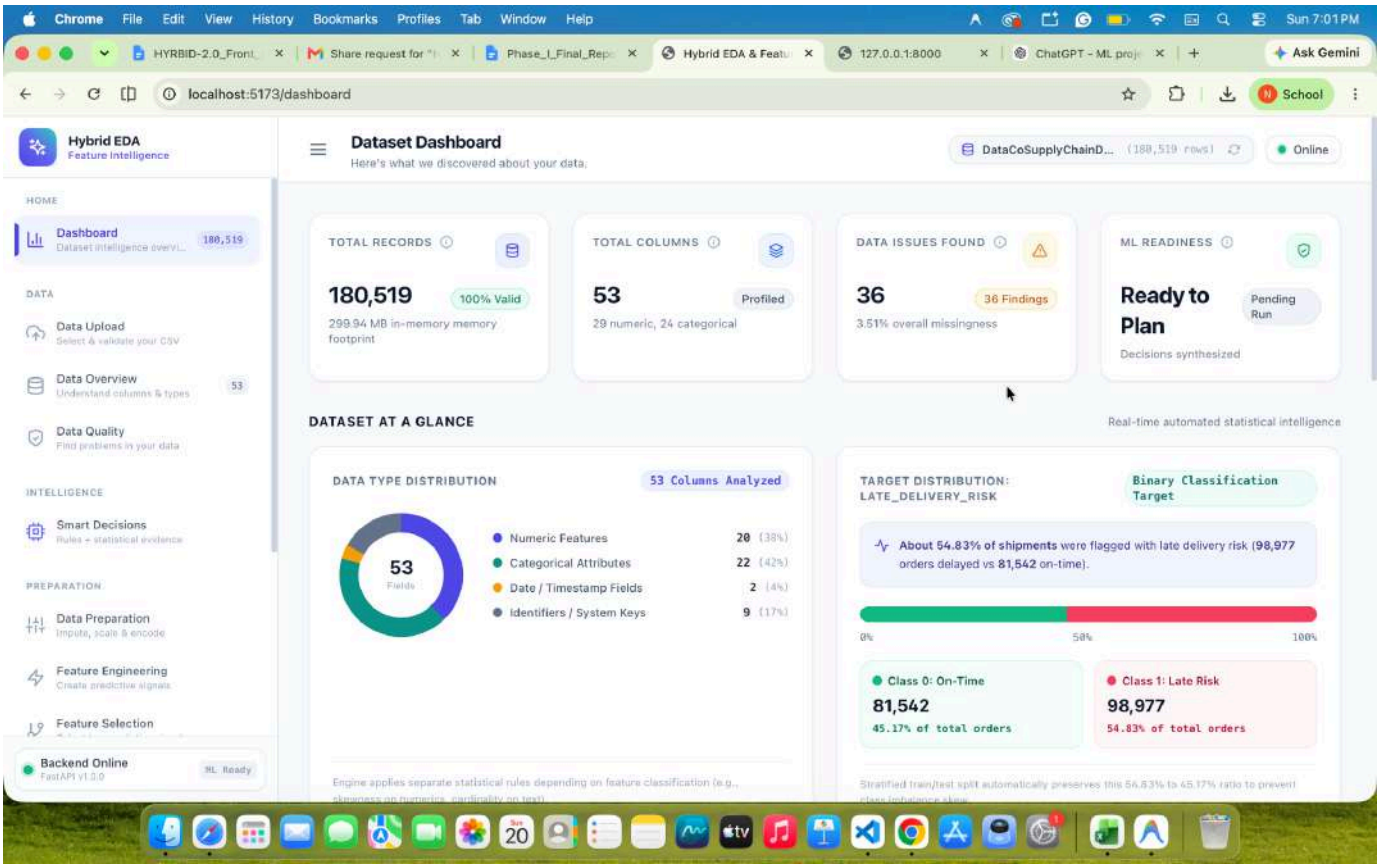
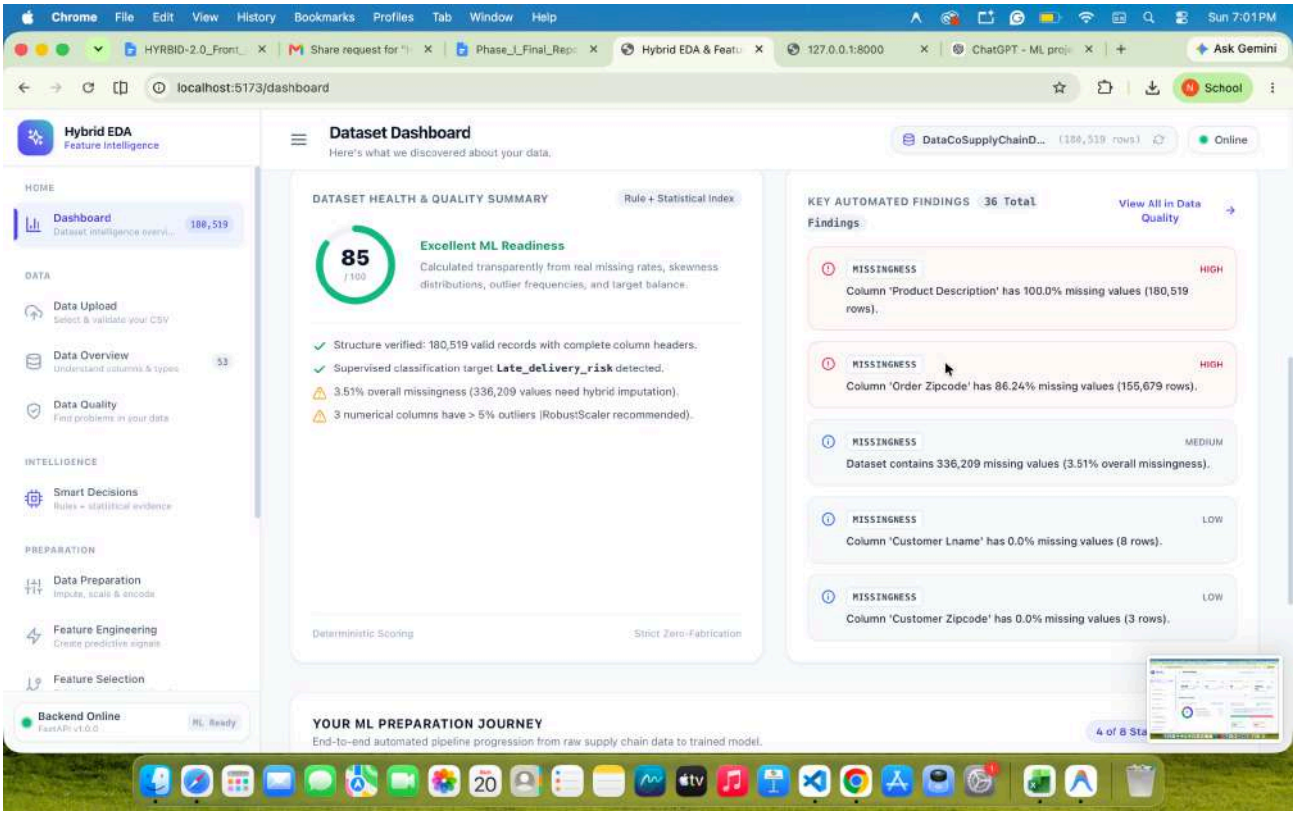
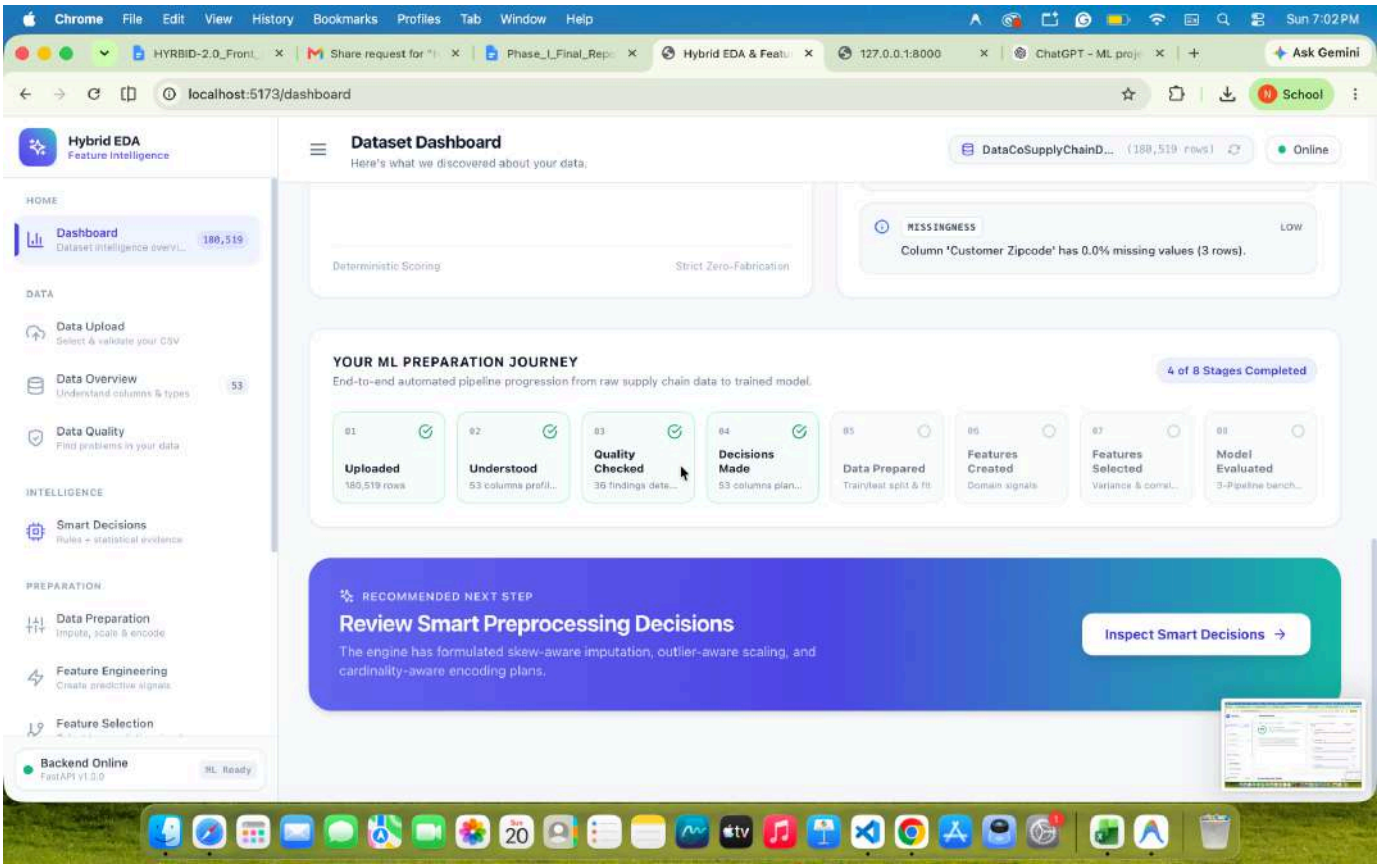
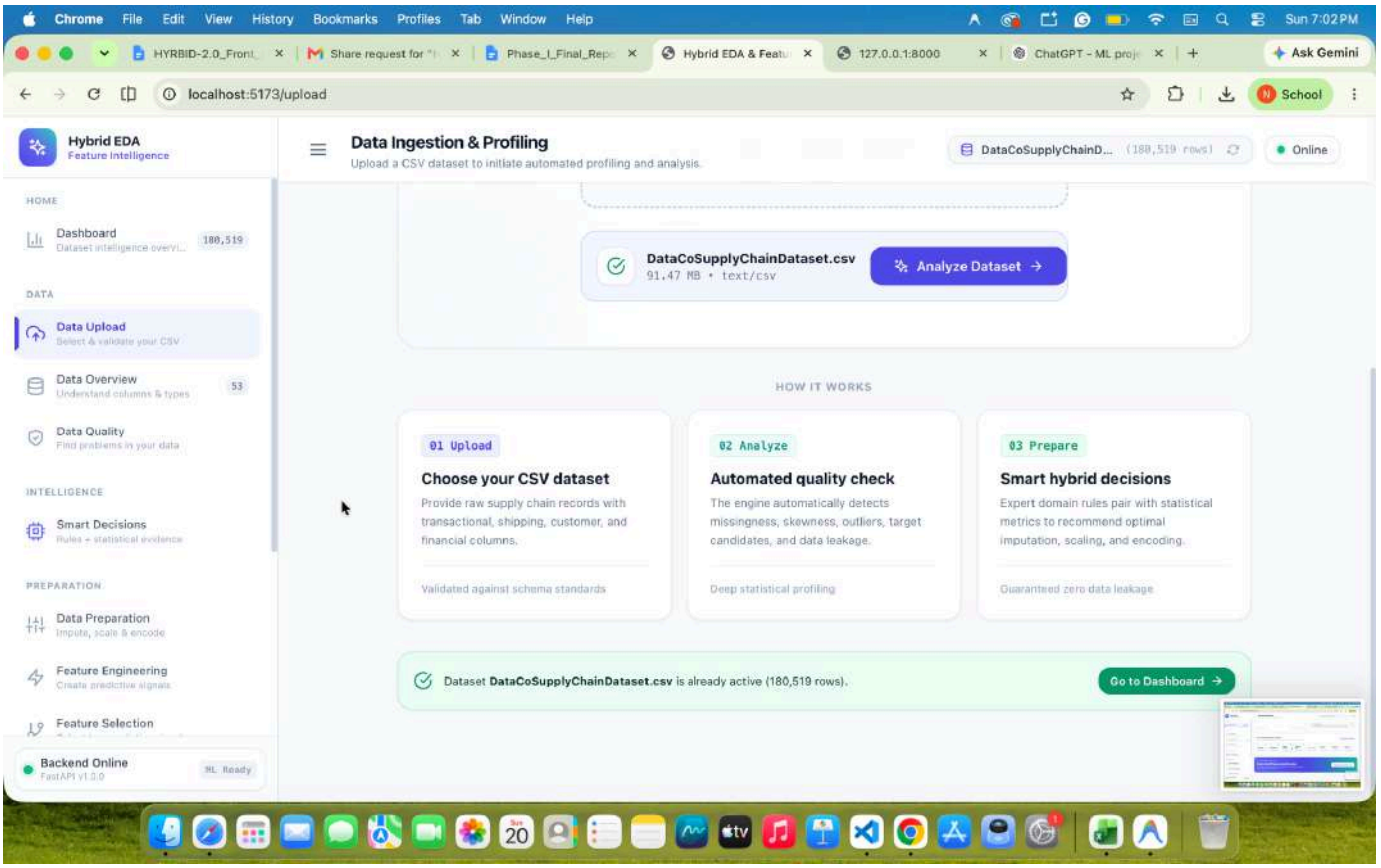
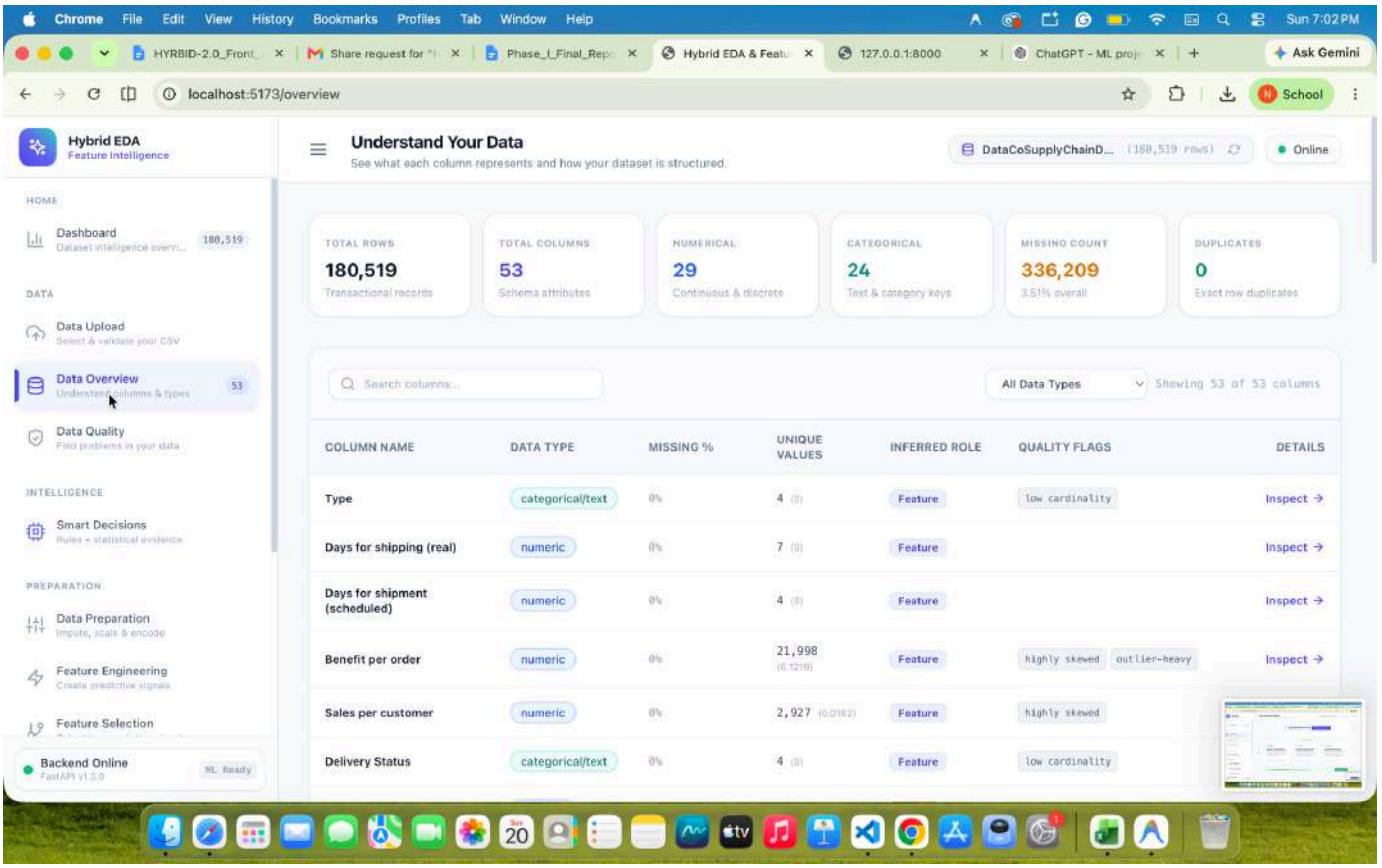
HYBRID EDA AND FEATURE ENGINEERING FRAMEWORK FOR SUPPLY CHAIN DATASET

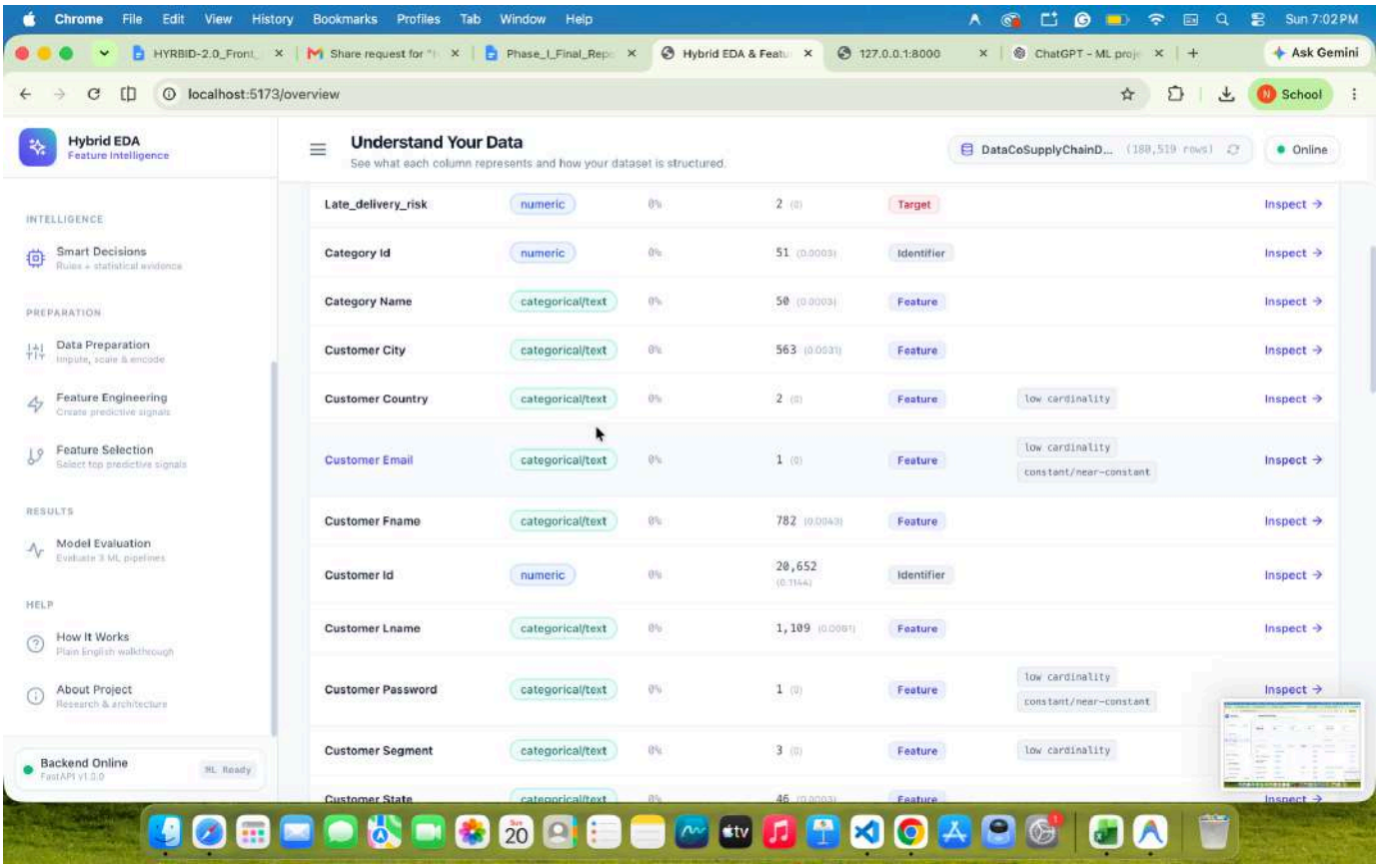


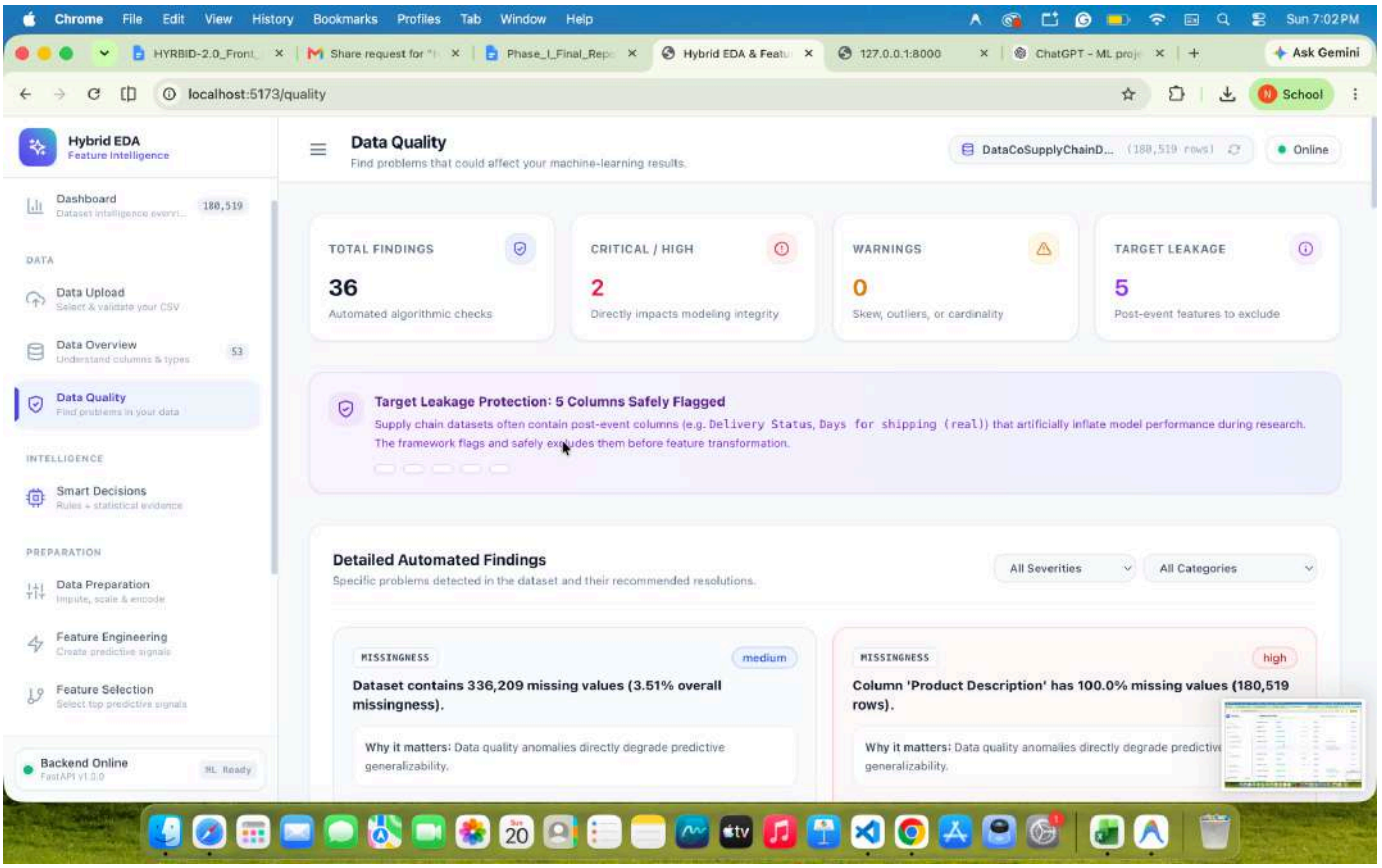


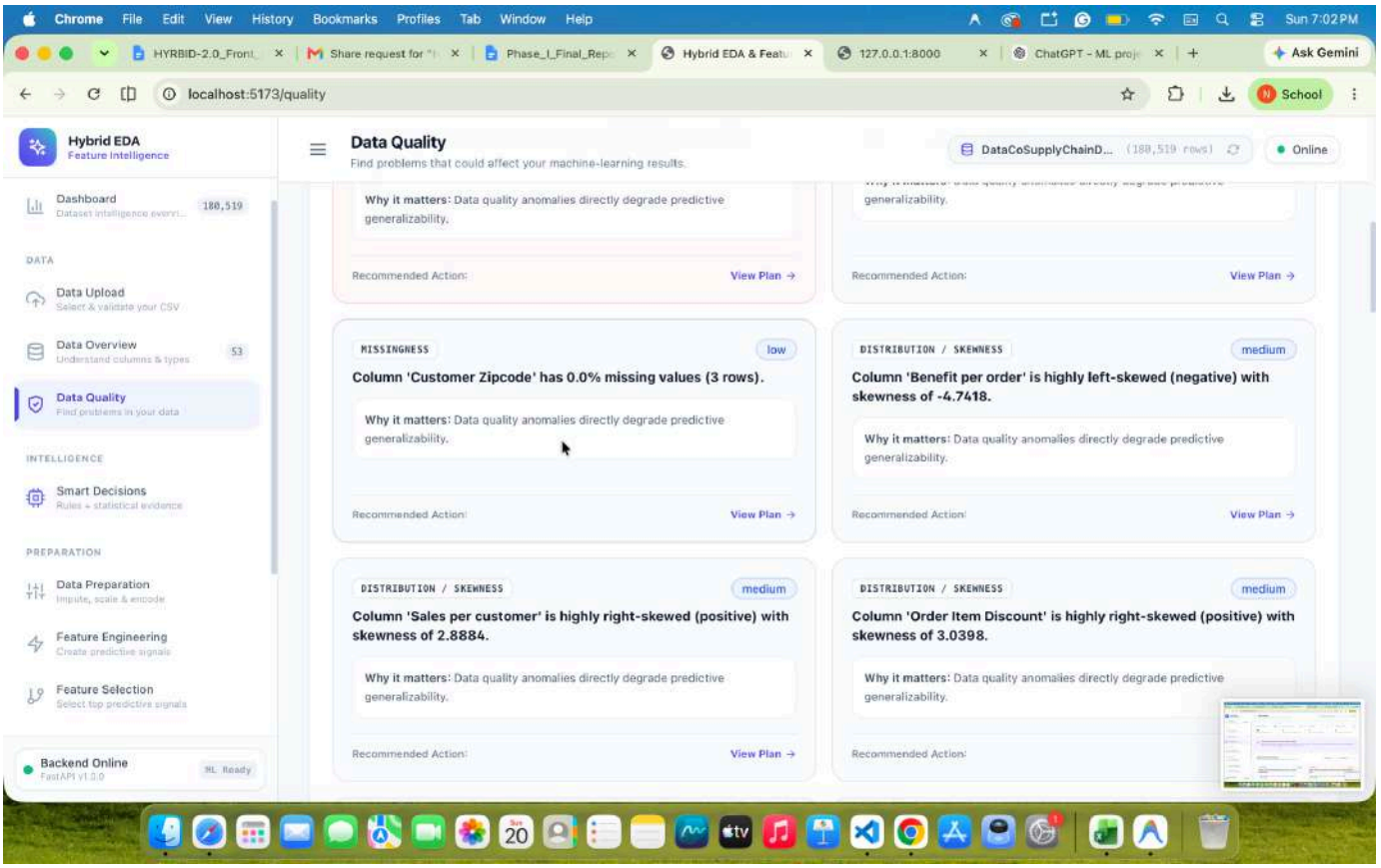

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/dashboard
Hybrid EDA
Feature Intelligence
Dataset Dashboard
Here's what we discovered about your data.
DataCoSupplyChainD... (180,519 rows)
Online
HOME
Dashboard
180,519
DATA
Data Upload
Data Overview
53
Data Quality
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Feature Engineering
Feature Selection
Backend Online
FastAPI v1.0.0
ML Ready
Deterministic Scoring
Strict Zero-Fabrication
MISSINGNESS
LOW
Column 'Customer Zipcode' has 0.0% missing values (3 rows).
YOUR ML PREPARATION JOURNEY
End-to-end automated pipeline progression from raw supply chain data to trained model.
4 of 8 Stages Completed
01 Uploaded 180,519 rows
02 Understood 53 columns profil...
03 Quality Checked 36 findings dete...
04 Decisions Made 53 columns plan...
05 Data Prepared Train/test split & fit
06 Features Created Domain signals
07 Features Selected Variance & corr...
08 Model Evaluated 3-Pipeline bench...
RECOMMENDED NEXT STEP
Review Smart Preprocessing Decisions
The engine has formulated skew-aware imputation, outlier-aware scaling, and cardinality-aware encoding plans.
Inspect Smart Decisions →

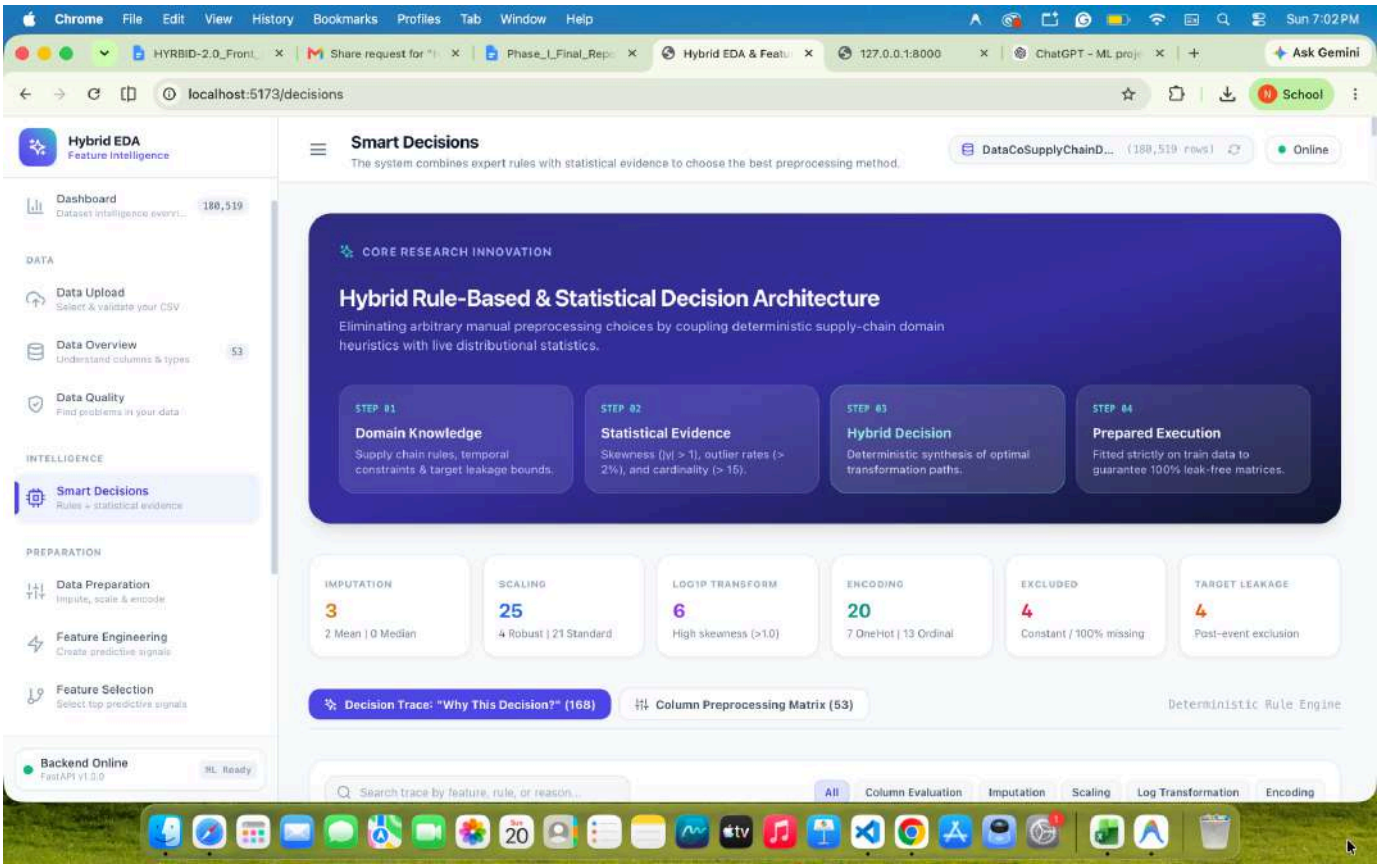

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/upload
Hybrid EDA
Feature Intelligence
Data Ingestion & Profiling
Upload a CSV dataset to initiate automated profiling and analysis.
DataCoSupplyChainD... (180,519 rows)
Online
HOME
Dashboard
180,519
DATA
Data Upload
Data Overview
53
Data Quality
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Feature Engineering
Feature Selection
Backend Online
FastAPI v1.0.0
ML Ready
DataCoSupplyChainDataset.csv
91.47 MB • text/csv
Analyze Dataset →
HOW IT WORKS
01 Upload
Choose your CSV dataset
Provide raw supply chain records with transactional, shipping, customer, and financial columns.
Validated against schema standards
02 Analyze
Automated quality check
The engine automatically detects missingness, skewness, outliers, target candidates, and data leakage.
Deep statistical profiling
03 Prepare
Smart hybrid decisions
Expert domain rules pair with statistical metrics to recommend optimal imputation, scaling, and encoding.
Guaranteed zero data leakage
Dataset DataCoSupplyChainDataset.csv is already active (180,519 rows).
Go to Dashboard →

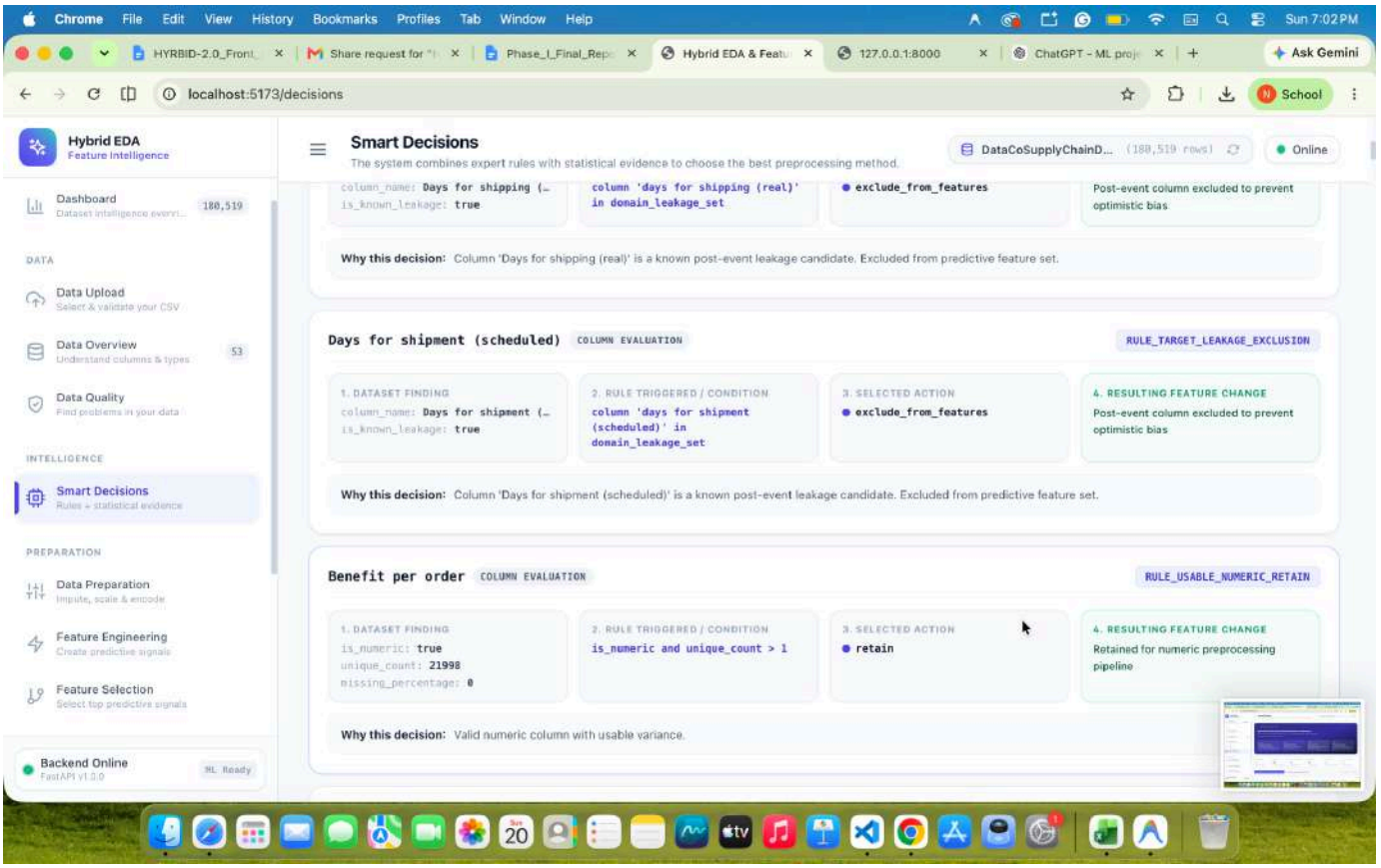

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/overview
Hybrid EDA
Feature Intelligence
Understand Your Data
See what each column represents and how your dataset is structured.
DataCoSupplyChainD... (180,519 rows)
Online
HOME
Dashboard
DATA
Data Upload
Data Overview
Data Quality
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Feature Engineering
Feature Selection
Backend Online
TOTAL ROWS
180,519
TOTAL COLUMNS
53
NUMERICAL
29
CATEGORICAL
24
MISSING COUNT
336,209
DUPLICATES
0
Search columns...
All Data Types
Showing 53 of 53 columns
COLUMN NAME
DATA TYPE
MISSING %
UNIQUE VALUES
INFERRED ROLE
QUALITY FLAGS
DETAILS
Type
categorical/text
Feature
low cardinality
Inspect
Days for shipping (real)
numeric
Days for shipment (scheduled)
Benefit per order
highly skewed
outlier-heavy
Sales per customer
Delivery Status

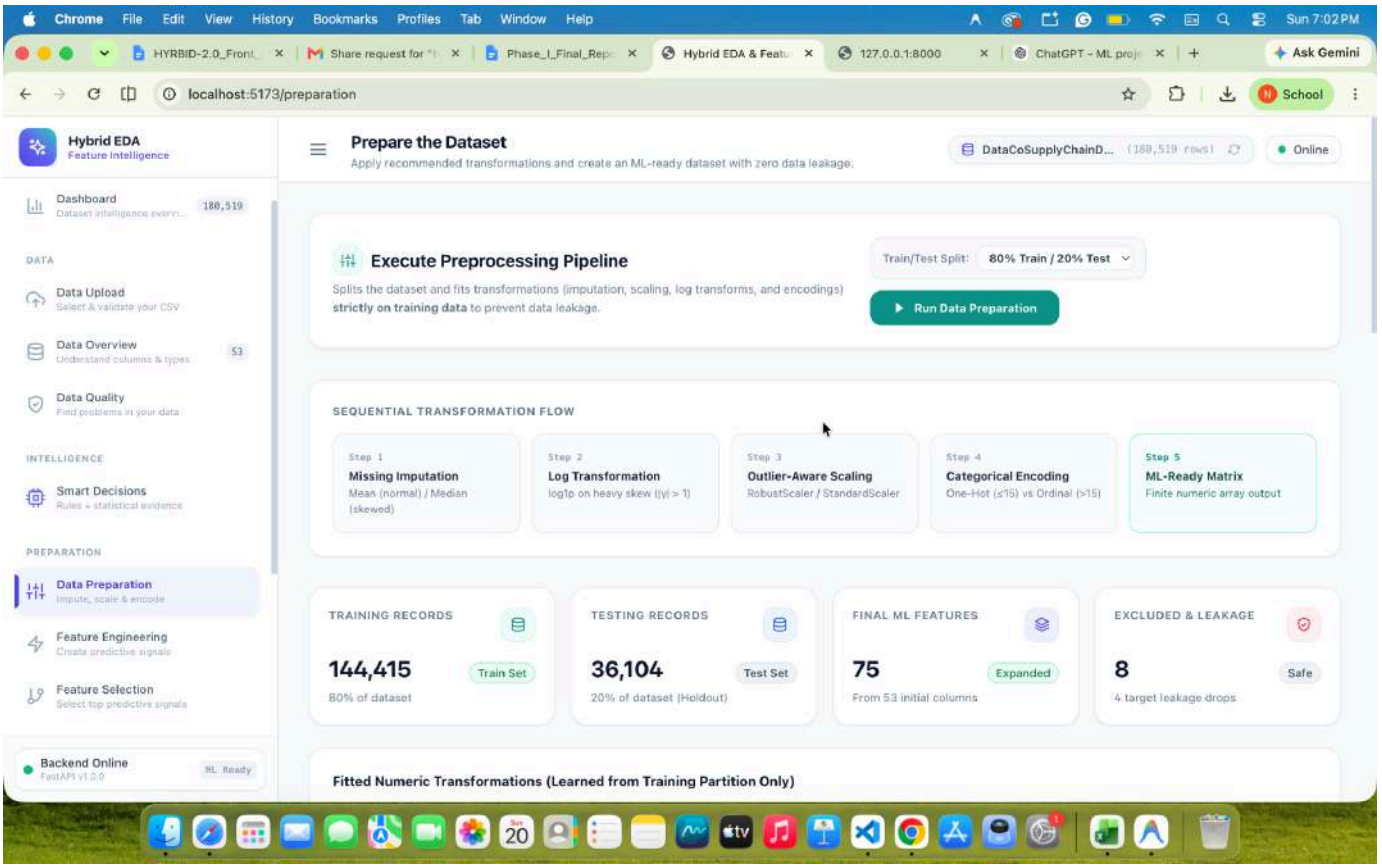

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/overview
Hybrid EDA
Feature Intelligence
Understand Your Data
See what each column represents and how your dataset is structured.
DataCoSupplyChainD... (180,519 rows)
Online
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Feature Engineering
Feature Selection
RESULTS
Model Evaluation
HELP
How It Works
About Project
Backend Online
Late_delivery_risk
numeric
Target
Category Id
Category Name
Customer City
Customer Country
Customer Email
Customer Fname
Customer Id
Customer Lname
Customer Password
Customer Segment
Customer State
low cardinality
constant/near-constant
Inspect

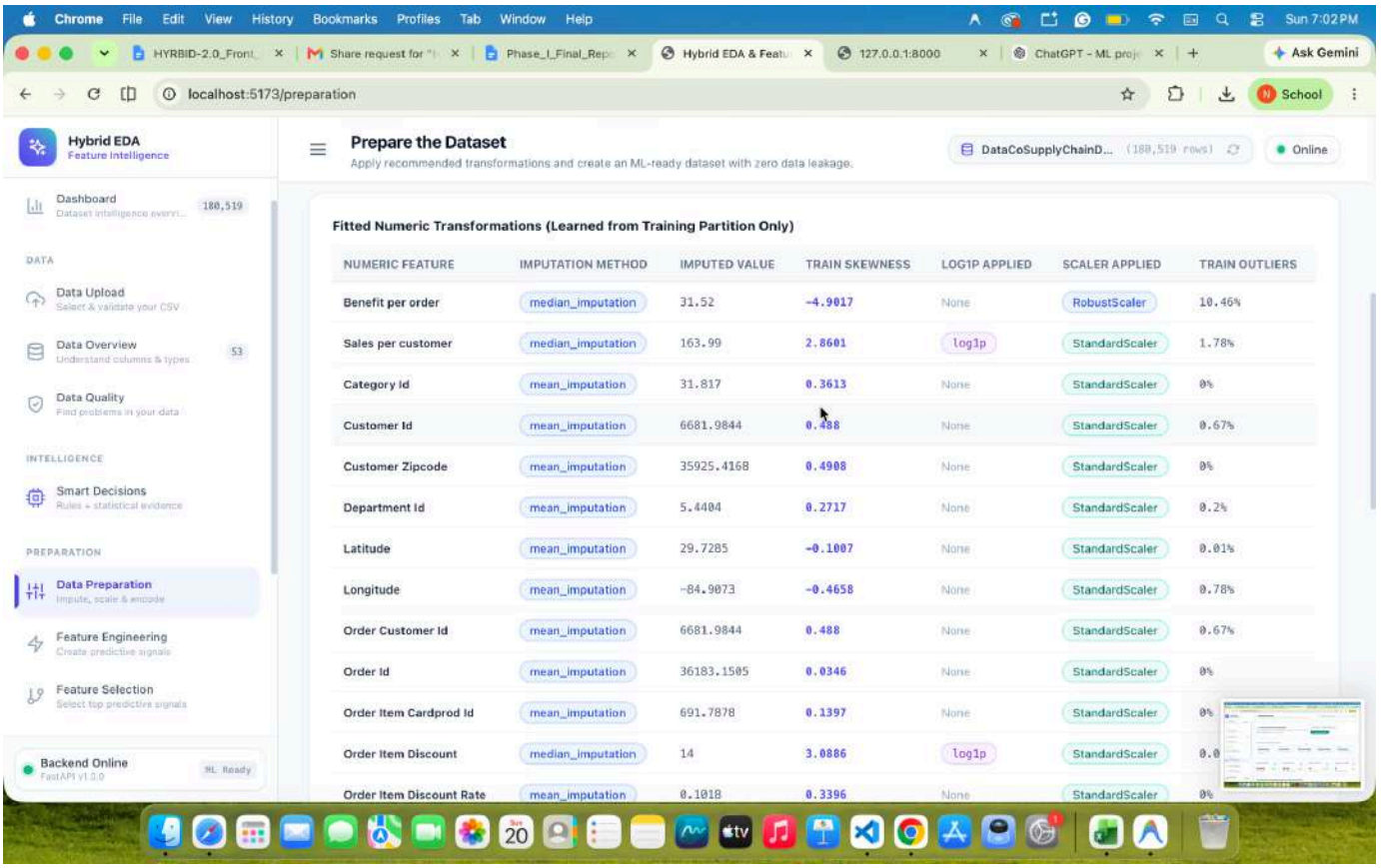

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/quality
Hybrid EDA
Feature Intelligence
Dashboard
180,519
DATA
Data Upload
Select & validate your CSV
Data Overview
53
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Feature Engineering
Feature Selection
Backend Online
Data Quality
Find problems that could affect your machine-learning results.
DataCoSupplyChainD... (180,519 rows)
Online
TOTAL FINDINGS
36
Automated algorithmic checks
CRITICAL / HIGH
2
Directly impacts modeling integrity
WARNINGS
0
Skew, outliers, or cardinality
TARGET LEAKAGE
5
Post-event features to exclude
Target Leakage Protection: 5 Columns Safely Flagged
Supply chain datasets often contain post-event columns (e.g. Delivery Status, Days for shipping (real)) that artificially inflate model performance during research.
The framework flags and safely excludes them before feature transformation.
Detailed Automated Findings
Specific problems detected in the dataset and their recommended resolutions.
All Severities
All Categories
MISSINGNESS
medium
Dataset contains 336,209 missing values (3.51% overall missingness).
Why it matters: Data quality anomalies directly degrade predictive generalizability.
MISSINGNESS
high
Column 'Product Description' has 100.0% missing values (180,519 rows).
Why it matters: Data quality anomalies directly degrade predictive generalizability.

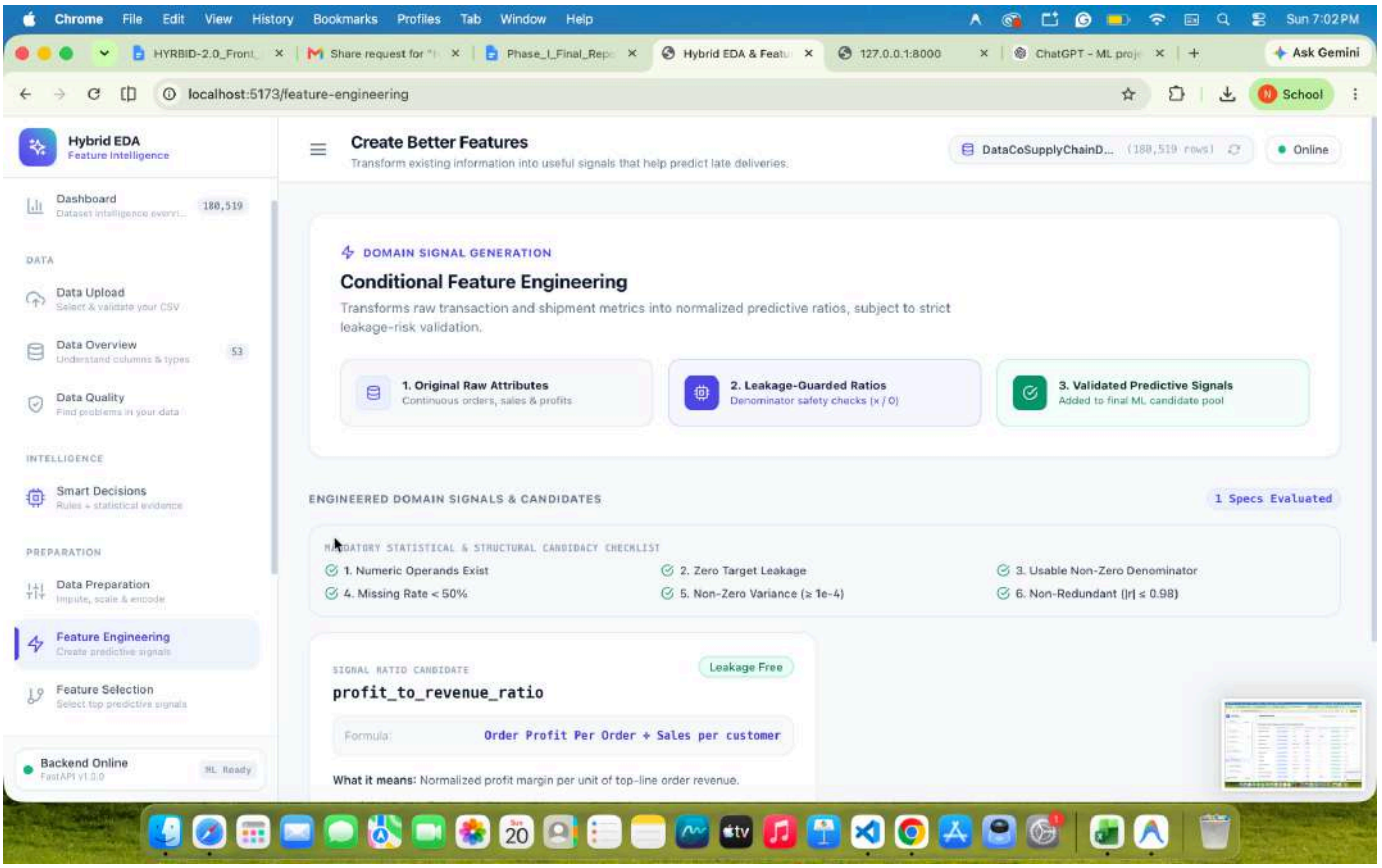

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/quality
Hybrid EDA
Feature Intelligence
Data Quality
Find problems that could affect your machine-learning results.
DataCoSupplyChainD... (180,519 rows)
Online
Why it matters: Data quality anomalies directly degrade predictive generalizability.
Recommended Action:
View Plan →
MISSINGNESS
low
Column 'Customer Zipcode' has 0.0% missing values (3 rows).
DISTRIBUTION / SKEWNESS
medium
Column 'Benefit per order' is highly left-skewed (negative) with skewness of -4.7418.
DISTRIBUTION / SKEWNESS
medium
Column 'Sales per customer' is highly right-skewed (positive) with skewness of 2.8884.
DISTRIBUTION / SKEWNESS
medium
Column 'Order Item Discount' is highly right-skewed (positive) with skewness of 3.0398.

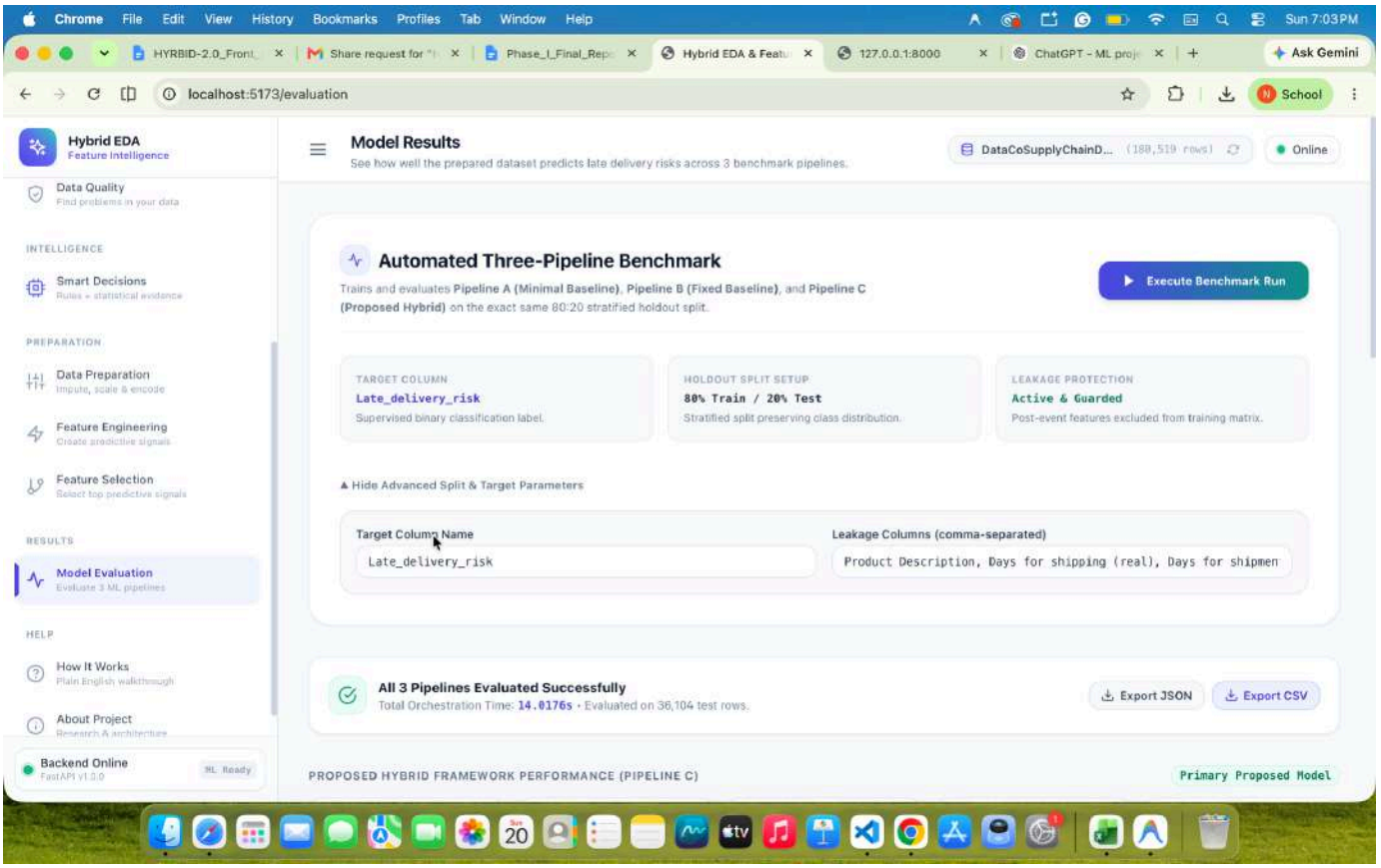

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/decisions
Hybrid EDA
Feature Intelligence
Smart Decisions
The system combines expert rules with statistical evidence to choose the best preprocessing method.
DataCoSupplyChainD... (180,519 rows)
Online
Dashboard
180,519
DATA
Data Upload
Select & validate your CSV
Data Overview
53
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
Rules + statistical evidence
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
Backend Online
FastAPI v1.0.0
ML Ready
CORE RESEARCH INNOVATION
Hybrid Rule-Based & Statistical Decision Architecture
Eliminating arbitrary manual preprocessing choices by coupling deterministic supply-chain domain heuristics with live distributional statistics.
STEP 01
Domain Knowledge
Supply chain rules, temporal constraints & target leakage bounds.
STEP 02
Statistical Evidence
Skewness (|γ| > 1), outlier rates (> 2%), and cardinality (> 15).
STEP 03
Hybrid Decision
Deterministic synthesis of optimal transformation paths.
STEP 04
Prepared Execution
Fitted strictly on train data to guarantee 100% leak-free matrices.
IMPUTATION
3
2 Mean | 0 Median
SCALING
25
4 Robust | 21 Standard
LOG1P TRANSFORM
6
High skewness (>1.0)
ENCODING
20
7 OneHot | 13 Ordinal
EXCLUDED
4
Constant / 100% missing
TARGET LEAKAGE
4
Post-event exclusion
Decision Trace: "Why This Decision?" (168)
Column Preprocessing Matrix (53)
Deterministic Rule Engine
Search trace by feature, rule, or reason...
All
Column Evaluation
Imputation
Scaling
Log Transformation
Encoding

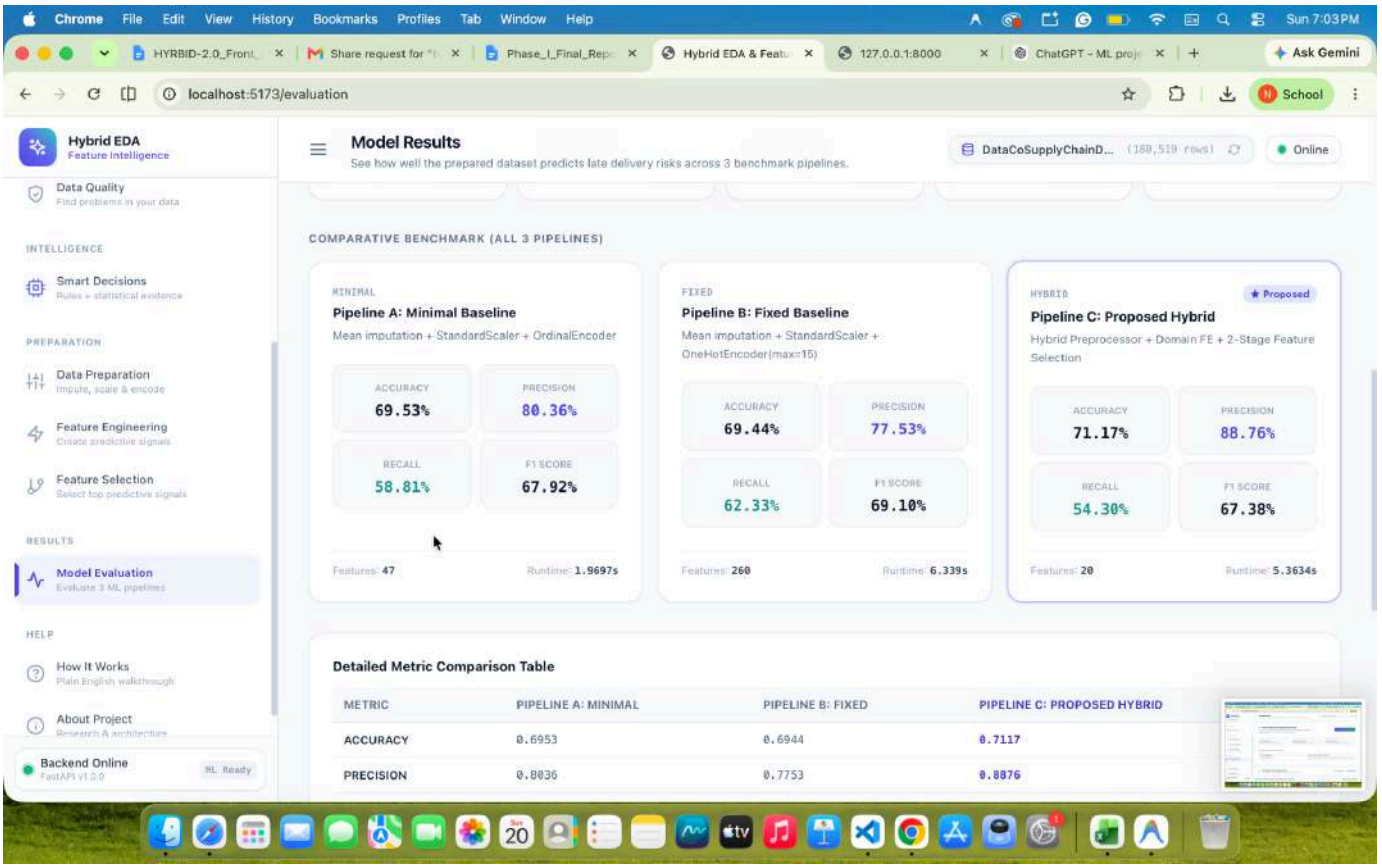

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/decisions
Hybrid EDA
Feature Intelligence
Smart Decisions
The system combines expert rules with statistical evidence to choose the best preprocessing method.
DataCoSupplyChainD... (180,519 rows)
Online
column_name: Days for shipping (...
is_known_leakage: true
column 'days for shipping (real)' in domain_leakage_set
exclude_from_features
Post-event column excluded to prevent optimistic bias
Why this decision: Column 'Days for shipping (real)' is a known post-event leakage candidate. Excluded from predictive feature set.
Days for shipment (scheduled)
COLUMN EVALUATION
RULE_TARGET_LEAKAGE_EXCLUSION
1. DATASET FINDING
column_name: Days for shipment (...
is_known_leakage: true
2. RULE TRIGGERED / CONDITION
column 'days for shipment (scheduled)' in domain_leakage_set
3. SELECTED ACTION
exclude_from_features
4. RESULTING FEATURE CHANGE
Post-event column excluded to prevent optimistic bias
Why this decision: Column 'Days for shipment (scheduled)' is a known post-event leakage candidate. Excluded from predictive feature set.
Benefit per order
COLUMN EVALUATION
RULE_USABLE_NUMERIC_RETAIN
1. DATASET FINDING
is_numeric: true
unique_count: 21998
missing_percentage: 0
2. RULE TRIGGERED / CONDITION
is_numeric and unique_count > 1
3. SELECTED ACTION
retain
4. RESULTING FEATURE CHANGE
Retained for numeric preprocessing pipeline
Why this decision: Valid numeric column with usable variance.

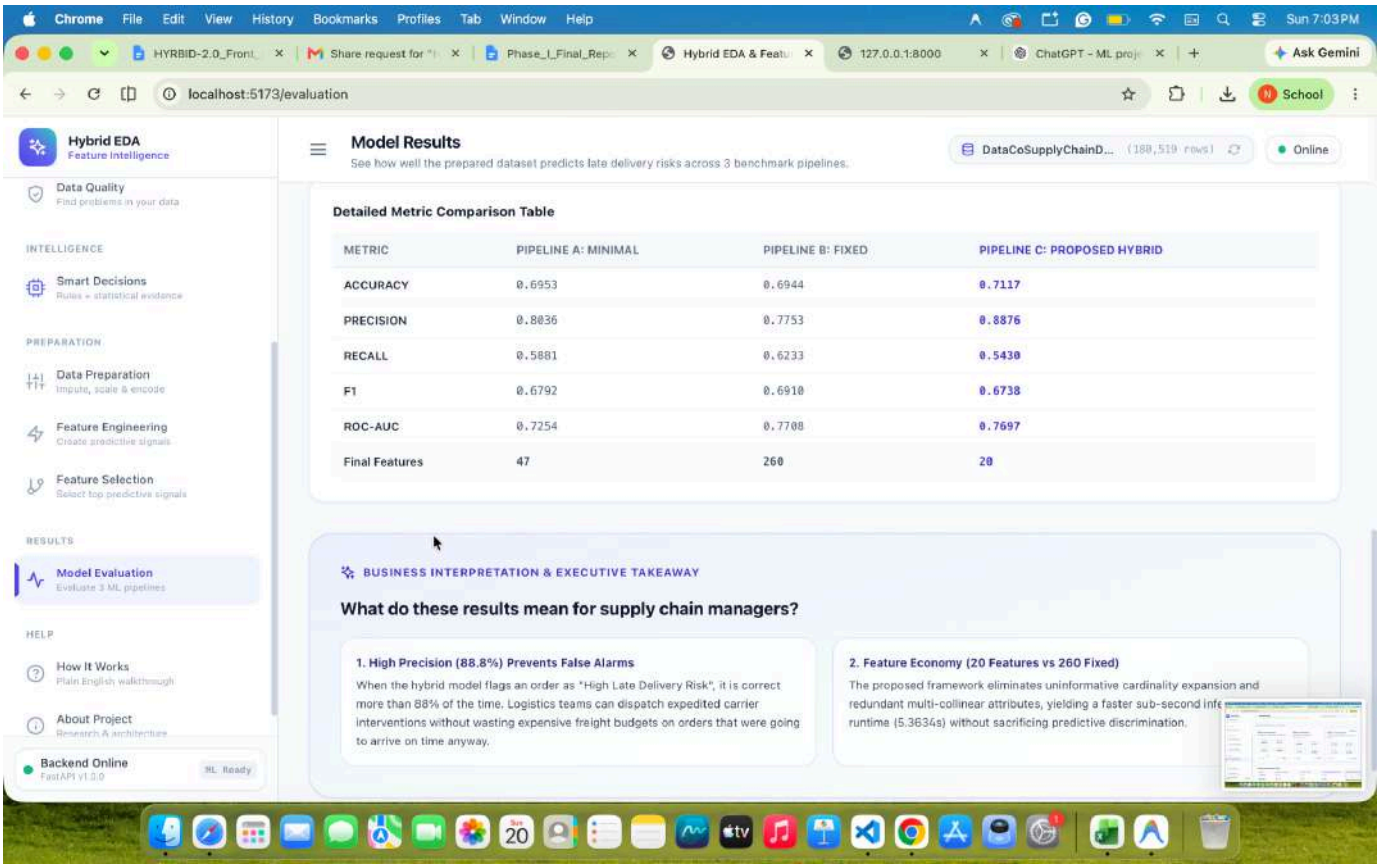

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/preparation
Hybrid EDA
Feature Intelligence
Dashboard
180,519
DATA
Data Upload
Select & validate your CSV
Data Overview
53
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
Rules + statistical evidence
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
Backend Online
FastAPI v1.0.0
ML Ready
Prepare the Dataset
Apply recommended transformations and create an ML-ready dataset with zero data leakage.
DataCoSupplyChainD... (180,519 rows)
Online
Execute Preprocessing Pipeline
Splits the dataset and fits transformations (imputation, scaling, log transforms, and encodings) strictly on training data to prevent data leakage.
Train/Test Split: 80% Train / 20% Test
Run Data Preparation
SEQUENTIAL TRANSFORMATION FLOW
Step 1
Missing Imputation
Mean (normal) / Median (skewed)
Step 2
Log Transformation
log1p on heavy skew (|γ| > 1)
Step 3
Outlier-Aware Scaling
RobustScaler / StandardScaler
Step 4
Categorical Encoding
One-Hot (≤15) vs Ordinal (>15)
Step 5
ML-Ready Matrix
Finite numeric array output
TRAINING RECORDS
144,415
Train Set
80% of dataset
TESTING RECORDS
36,104
Test Set
20% of dataset (Holdout)
FINAL ML FEATURES
75
Expanded
From 53 initial columns
EXCLUDED & LEAKAGE
8
Safe
4 target leakage drops
Fitted Numeric Transformations (Learned from Training Partition Only)

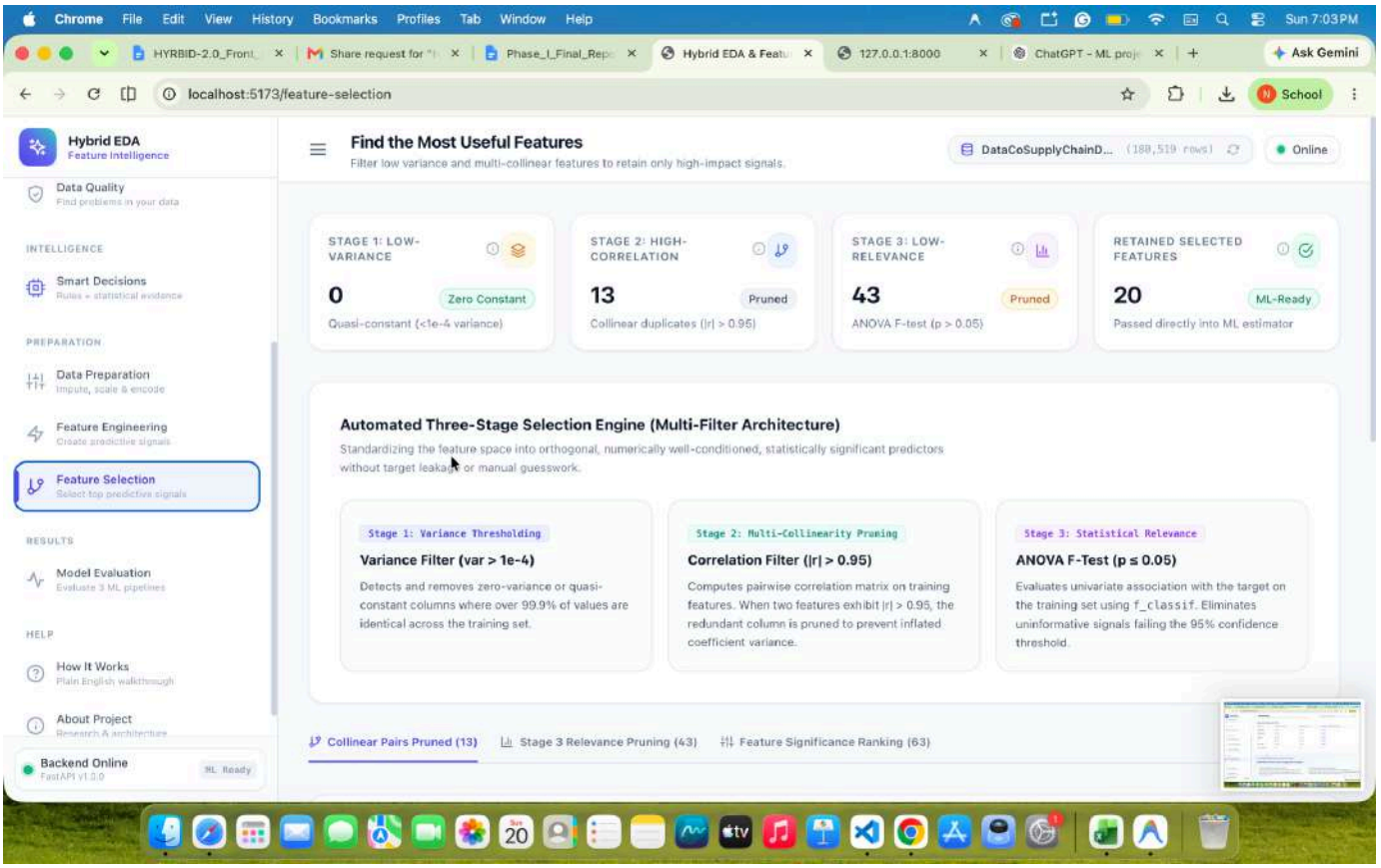

Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/preparation
Hybrid EDA
Feature Intelligence
Prepare the Dataset
Apply recommended transformations and create an ML-ready dataset with zero data leakage.
DataCoSupplyChainD... (180,519 rows)
Online
Fitted Numeric Transformations (Learned from Training Partition Only)
NUMERIC FEATURE | IMPUTATION METHOD | IMPUTED VALUE | TRAIN SKEWNESS | LOG1P APPLIED | SCALER APPLIED | TRAIN OUTLIERS
Benefit per order | median_imputation | 31.52 | -4.9017 | None | RobustScaler | 10.46%
Sales per customer | median_imputation | 163.99 | 2.8601 | log1p | StandardScaler | 1.78%
Category Id | mean_imputation | 31.817 | 0.3613 | None | StandardScaler | 0%
Customer Id | mean_imputation | 6681.9844 | 0.488 | None | StandardScaler | 0.67%
Customer Zipcode | mean_imputation | 35925.4168 | 0.4908 | None | StandardScaler | 0%
Department Id | mean_imputation | 5.4404 | 0.2717 | None | StandardScaler | 0.2%
Latitude | mean_imputation | 29.7285 | -0.1007 | None | StandardScaler | 0.01%
Longitude | mean_imputation | -84.9073 | -0.4658 | None | StandardScaler | 0.78%
Order Customer Id | mean_imputation | 6681.9844 | 0.488 | None | StandardScaler | 0.67%
Order Id | mean_imputation | 36183.1505 | 0.0346 | None | StandardScaler | 0%
Order Item Cardprod Id | mean_imputation | 691.7878 | 0.1397 | None | StandardScaler | 0%
Order Item Discount | median_imputation | 14 | 3.0886 | log1p | StandardScaler | 0.0
Order Item Discount Rate | mean_imputation | 0.1018 | 0.3396 | None | StandardScaler | 0%


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:02 PM
localhost:5173/feature-engineering
Hybrid EDA
Feature Intelligence
Dashboard
180,519
DATA
Data Upload
Select & validate your CSV
Data Overview
53
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Backend Online
ML Ready
Create Better Features
Transform existing information into useful signals that help predict late deliveries.
DataCoSupplyChainD... (180,519 rows)
Online
DOMAIN SIGNAL GENERATION
Conditional Feature Engineering
Transforms raw transaction and shipment metrics into normalized predictive ratios, subject to strict leakage-risk validation.
1. Original Raw Attributes
Continuous orders, sales & profits
2. Leakage-Guarded Ratios
Denominator safety checks (x / 0)
3. Validated Predictive Signals
Added to final ML candidate pool
ENGINEERED DOMAIN SIGNALS & CANDIDATES
1 Specs Evaluated
1. Numeric Operands Exist
2. Zero Target Leakage
3. Usable Non-Zero Denominator
4. Missing Rate < 50%
5. Non-Zero Variance (≥ 1e-4)
6. Non-Redundant (|r| ≤ 0.98)
SIGNAL RATIO CANDIDATE
Leakage Free
profit_to_revenue_ratio
Formula:
Order Profit Per Order ÷ Sales per customer
What it means: Normalized profit margin per unit of top-line order revenue.


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:03 PM
localhost:5173/evaluation
Hybrid EDA
Feature Intelligence
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
RESULTS
Model Evaluation
Evaluate 3 ML pipelines
HELP
How It Works
Plain English walkthrough
About Project
Backend Online
ML Ready
Model Results
See how well the prepared dataset predicts late delivery risks across 3 benchmark pipelines.
DataCoSupplyChainD... (180,519 rows)
Online
Automated Three-Pipeline Benchmark
Trains and evaluates Pipeline A (Minimal Baseline), Pipeline B (Fixed Baseline), and Pipeline C (Proposed Hybrid) on the exact same 80:20 stratified holdout split.
Execute Benchmark Run
TARGET COLUMN
Late_delivery_risk
Supervised binary classification label.
HOLDOUT SPLIT SETUP
80% Train / 20% Test
Stratified split preserving class distribution.
LEAKAGE PROTECTION
Active & Guarded
Post-event features excluded from training matrix.
▲ Hide Advanced Split & Target Parameters
Target Column Name
Late_delivery_risk
Leakage Columns (comma-separated)
Product Description, Days for shipping (real), Days for shipmen
All 3 Pipelines Evaluated Successfully
Total Orchestration Time: 14.0176s · Evaluated on 36,104 test rows.
Export JSON
Export CSV
PROPOSED HYBRID FRAMEWORK PERFORMANCE (PIPELINE C)
Primary Proposed Model


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:03 PM
localhost:5173/evaluation
Hybrid EDA
Feature Intelligence
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
RESULTS
Model Evaluation
Evaluate 3 ML pipelines
HELP
How It Works
Plain English walkthrough
About Project
Research & architecture
Backend Online
FastAPI v1.0.0
ML Ready
Model Results
See how well the prepared dataset predicts late delivery risks across 3 benchmark pipelines.
DataCoSupplyChainD... (180,519 rows)
Online
COMPARATIVE BENCHMARK (ALL 3 PIPELINES)
MINIMAL
Pipeline A: Minimal Baseline
Mean imputation + StandardScaler + OrdinalEncoder
ACCURACY 69.53%
PRECISION 80.36%
RECALL 58.81%
F1 SCORE 67.92%
Features: 47
Runtime: 1.9697s
FIXED
Pipeline B: Fixed Baseline
Mean imputation + StandardScaler + OneHotEncoder(max=15)
ACCURACY 69.44%
PRECISION 77.53%
RECALL 62.33%
F1 SCORE 69.10%
Features: 260
Runtime: 6.339s
HYBRID
★ Proposed
Pipeline C: Proposed Hybrid
Hybrid Preprocessor + Domain FE + 2-Stage Feature Selection
ACCURACY 71.17%
PRECISION 88.76%
RECALL 54.30%
F1 SCORE 67.38%
Features: 20
Runtime: 5.3634s
Detailed Metric Comparison Table
METRIC
PIPELINE A: MINIMAL
PIPELINE B: FIXED
PIPELINE C: PROPOSED HYBRID
ACCURACY
0.6953
0.6944
0.7117
PRECISION
0.8036
0.7753
0.8876


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:03 PM
localhost:5173/evaluation
Hybrid EDA
Feature Intelligence
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
RESULTS
Model Evaluation
Evaluate 3 ML pipelines
HELP
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About Project
Research & architecture
Backend Online
FastAPI v1.0.0
ML Ready
Model Results
See how well the prepared dataset predicts late delivery risks across 3 benchmark pipelines.
DataCoSupplyChainD... (180,519 rows)
Online
Detailed Metric Comparison Table
METRIC
PIPELINE A: MINIMAL
PIPELINE B: FIXED
PIPELINE C: PROPOSED HYBRID
ACCURACY
0.6953
0.6944
0.7117
PRECISION
0.8036
0.7753
0.8876
RECALL
0.5881
0.6233
0.5430
F1
0.6792
0.6910
0.6738
ROC-AUC
0.7254
0.7708
0.7697
Final Features
47
260
20
BUSINESS INTERPRETATION & EXECUTIVE TAKEAWAY
What do these results mean for supply chain managers?
1. High Precision (88.8%) Prevents False Alarms
When the hybrid model flags an order as "High Late Delivery Risk", it is correct more than 88% of the time. Logistics teams can dispatch expedited carrier interventions without wasting expensive freight budgets on orders that were going to arrive on time anyway.
2. Feature Economy (20 Features vs 260 Fixed)
The proposed framework eliminates uninformative cardinality expansion and redundant multi-collinear attributes, yielding a faster sub-second inf
runtime (5.3634s) without sacrificing predictive discrimination.


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sun 7:03 PM
localhost:5173/feature-selection
Hybrid EDA
Feature Intelligence
Data Quality
Find problems in your data
INTELLIGENCE
Smart Decisions
PREPARATION
Data Preparation
Impute, scale & encode
Feature Engineering
Create predictive signals
Feature Selection
Select top predictive signals
RESULTS
Model Evaluation
Evaluate 3 ML pipelines
HELP
How It Works
Plain English walkthrough
About Project
Research & architecture
Backend Online
FastAPI v1.0.0
ML Ready
Find the Most Useful Features
Filter low variance and multi-collinear features to retain only high-impact signals.
DataCoSupplyChainD... (180,519 rows)
Online
STAGE 1: LOW-VARIANCE
0
Zero Constant
Quasi-constant (<1e-4 variance)
STAGE 2: HIGH-CORRELATION
13
Pruned
Collinear duplicates (|r| > 0.95)
STAGE 3: LOW-RELEVANCE
43
Pruned
ANOVA F-test (p > 0.05)
RETAINED SELECTED FEATURES
20
ML-Ready
Passed directly into ML estimator
Automated Three-Stage Selection Engine (Multi-Filter Architecture)
Standardizing the feature space into orthogonal, numerically well-conditioned, statistically significant predictors without target leakage or manual guesswork.
Stage 1: Variance Thresholding
Variance Filter (var > 1e-4)
Detects and removes zero-variance or quasi-constant columns where over 99.9% of values are identical across the training set.
Stage 2: Multi-Collinearity Pruning
Correlation Filter (|r| > 0.95)
Computes pairwise correlation matrix on training features. When two features exhibit |r| > 0.95, the redundant column is pruned to prevent inflated coefficient variance.
Stage 3: Statistical Relevance
ANOVA F-Test (p ≤ 0.05)
Evaluates univariate association with the target on the training set using f_classif. Eliminates uninformative signals failing the 95% confidence threshold.
Collinear Pairs Pruned (13)
Stage 3 Relevance Pruning (43)
Feature Significance Ranking (63)